Phase 2 Report

CS 5542 – Snowflake-Integrated Financial Analytics System

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1. Dataset & Knowledge Base Documentation

1.1 Dataset Name, Modality, and Source

The dataset used in this project consists of historical Toyota stock price data spanning from 1980 to 2026. The data is stored in CSV format under the data/ directory of the repository.

The dataset includes structured financial information such as:

- Date
- Open price
- High price
- Low price
- Close price
- Volume

The knowledge base for this system is structured and tabular, focusing on time-series financial data rather than unstructured documents. The dataset is locally stored and ingested into Snowflake for warehouse-level processing and analytics.

1.2 Domain Relevance

The dataset supports the development of a reproducible cloud-integrated analytics pipeline. Financial time-series data provides a clear use case for:

- Structured data ingestion
- Data warehousing
- SQL-based analytics

- Application-layer query execution
- Logging and reproducibility

This domain allows demonstration of a complete end-to-end pipeline integrating local data sources with a cloud data warehouse and an interactive application interface.

1.3 Data Ingestion and Preprocessing

The CSV dataset is processed using Python ingestion scripts located in the scripts/ directory.

The ingestion workflow includes:

- Establishing a connection to Snowflake
- Creating database and schema structures
- Defining structured tables
- Uploading the CSV file to a Snowflake stage
- Loading staged data into warehouse tables

Basic preprocessing ensures correct schema mapping between the CSV file and Snowflake tables, including column type alignment and data validation.

No sampling was performed. The full dataset was ingested to preserve historical completeness and enable comprehensive analysis.

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2. Processing & Analytics Pipeline

2.1 Warehouse-Based Processing

The project implements a warehouse-driven analytics architecture rather than a document-retrieval pipeline.

The core processing flow is:

Local CSV → Snowflake Stage → Snowflake Table → SQL Queries → Streamlit Application

All SQL scripts required for schema creation and data loading are included in the repository.

2.2 Snowflake Schema

The Snowflake schema is defined using SQL scripts and includes:

- A structured stock price table
- Logging support for application queries
- Tables designed for analytical operations

The schema enables structured querying of historical stock prices and supports aggregation, filtering, and time-based analysis.

2.3 Analytical Query Execution

Example analytical queries include:

- Filtering stock prices by date range
- Computing average closing prices
- Identifying historical price trends
- Evaluating price volatility patterns

These queries validate warehouse connectivity and demonstrate successful ingestion and structured analytics.

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3. Application Integration

3.1 Interface

The project includes a Streamlit-based application that connects directly to Snowflake.

The application enables users to:

- Establish a Snowflake connection
- Execute analytical SQL queries
- View structured stock price outputs

The interface serves as the presentation layer for the warehouse-integrated analytics system.

3.2 Evidence Display

Query results are displayed directly within the Streamlit interface in structured tabular format.

The outputs reflect real-time results retrieved from Snowflake tables, ensuring that all displayed information is grounded in warehouse data.

3.3 Query Logging

User interactions and executed queries are logged to a structured logging file under the logs/ directory.

This logging mechanism supports:

- Reproducibility
- Auditability
- Future warehouse-level logging integration

Logging ensures traceability of executed queries and system behavior.

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4. Snowflake Data Pipeline & Schema

4.1 Schema Definition

Database and schema creation are defined in SQL scripts included in the repository. These scripts create the required tables for storing historical stock price data.

The schema ensures consistent column structure and supports time-series analysis operations.

4.2 Data Ingestion and Integration

The ingestion pipeline:

1. Stages the local CSV file in Snowflake

2. Loads the staged file into structured tables
3. Validates successful data load

Python scripts and SQL files together provide a fully reproducible ingestion process.

The Streamlit application connects directly to Snowflake to execute queries and retrieve analytical results.

This ensures a seamless integration between warehouse storage and application interface.

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5. Reproducibility Plan

5.1 Environment

All required dependencies are specified in requirements.txt.

Core libraries include:

- streamlit
- snowflake-connector-python
- pandas
- numpy

This ensures that the environment can be recreated on any machine.

5.2 Configuration

Snowflake connection parameters are defined within the project's connection module. SQL scripts required for schema creation and data ingestion are included in the repository.

The dataset is stored under the data/ directory to ensure full reproducibility.

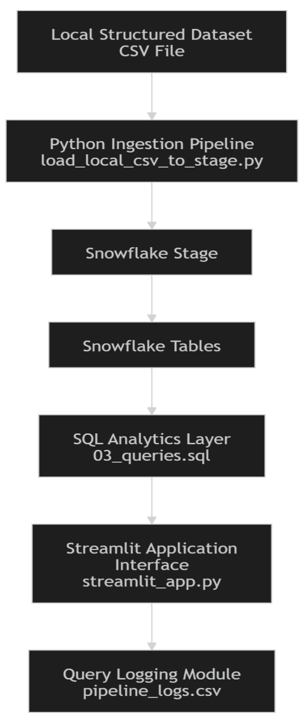
5.3 Run Instructions

To reproduce the system:

1. Clone the repository
2. Install dependencies using pip install -r requirements.txt
3. Configure Snowflake credentials
4. Execute SQL schema and ingestion scripts
5. Launch the Streamlit application

All scripts and configuration files necessary for reproduction are included in the repository.

System Architecture Summary



This architecture demonstrates a complete, reproducible, warehouse-integrated analytics system.